

Jayanth Dasamantharao

Masters of Science - Data Science

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AI Engineer - EliteUS Software Solutions • Edison, NJ

PROFESSIONAL SUMMARY

I am a Machine Learning Engineer and Data Scientist with over three years of combined industry and research experience, specializing in Large Language Models (LLMs), multi-agent AI systems, retrieval-augmented generation (RAG), deep learning, computer vision, and scalable ML system deployment. My work focuses on building **real-world, production-grade AI systems** that integrate seamlessly with enterprise workflows, optimize business processes, and deliver measurable improvements in efficiency, decision-making, and user engagement.

In my current role as an AI Engineer at EliteUS Software Solutions, I architect and implement end-to-end LLM-based applications—including multi-agent chatbot frameworks, intelligent routing engines, multimodal product assistants, and automated SQL generation pipelines. I work extensively with AWS Bedrock, Azure OpenAI, Hugging Face Transformers, AutoGen, Pinecone, ChromaDB, and LangChain to build robust RAG pipelines, inference-optimized deployments, and context-aware reasoning agents. My responsibilities span from **model evaluation, fine-tuning, and prompt optimization** to **vector database design, data ingestion pipelines, and scaling inference services**. I also develop CLIP-based similarity systems, Florence-powered CV workflows, and autonomous agent architectures that can validate queries, orchestrate tasks, and interact with user-facing applications in real-time.

Beyond core model development, I contribute heavily to system architecture and MLOps—deploying models through secure API gateways, integrating CI/CD (Git, OneDev), optimizing latency through quantization and batching, and implementing HAProxy for secure, fault-tolerant inference routing. I work closely with product and engineering teams to translate business needs into technical specifications, ensuring that each AI service is reliable, interpretable, and optimized for production-scale usage. My work directly supports enterprise clients by enabling automated help-desk operations, real-time product recommendations, intelligent workflow automation, and multimodal content understanding.

Prior to this, I served as a Data Science Intern at Harvest Software Solutions, where I built ML models to improve ad campaign forecasting accuracy by 70% and deployed sentiment analysis pipelines over multimillion-row datasets using FastAPI and Hugging Face. I also designed dashboards in Tableau and Power BI to present insights that guided strategic advertising decisions.

At Accenture (Regeneron Pharmaceuticals project), I gained deep experience in data engineering and large-scale data warehousing—optimizing ETL pipelines, automating analytics workflows, and improving system reliability through SQL performance tuning and AWS-based data integration. This background allows me to operate effectively at the intersection of **data infrastructure and applied machine learning**, enabling smooth end-to-end deployment of AI systems.

I hold an M.S. in Data Science from Rutgers University, where I conducted research in Bayesian modeling, NLP, fraud detection, and computer vision. My academic work spans Bayesian logistic regression (PyMC3), DistilBERT-based language identification, GloVe embeddings, Random Forests, XGBoost, CNNs, and real-time simulation environments for autonomous driving.

Outside of structured work, I am deeply committed to exploring frontiers in **agentic LLMs, multimodal representation learning, scalable RAG architectures, self-supervised learning, and reinforcement learning for autonomous reasoning**. My long-term goal is to build AI systems that exhibit robust reasoning, interpretability, and real-world reliability.

TECHNICAL SKILLS

Programming Languages	Python (Problem-solving) C++ MATLAB R (Statistics) SAS Scala.
Machine Learning (ML)	Regression Models Classification Models XGBoost Gradient Boosting Methods Ensemble Models Support Vector Machine Time Series Analysis.
Deep Learning (DL)	BERT NLP Large Language Models Word/Image Embedding Techniques LSTMs, GRU Gen AI Retrieval-Augmented Generation Langchain Prompt Tuning GPU Architecture.
Computer Vision	Object detection Image classification Semantic segmentation Neural Networks ADAS, Data Augmentation Vision Transformers Image Registration.
ML and DL Tools	Pytorch TensorFlow Keras HuggingFace OpenCV NLTK Spacy Numpy Pandas Matplotlib Seaborn scikit-learn/sklearn Scipy CUDA XGBoost H2O.ai Plotly.
Data Engineering & cloud	SQL PostgreSQL PySpark Hive Hadoop Azure Amazon Web Services Google Cloud Platform Data Warehousing Kubernetes MLOps Airflow Robotic Process Automation.
Data Visualization & other tools	Tableau Power BI Hypothesis testing Microsoft Excel (Vlookups, Power Pivot, Macros & VBA) JIRA CUDA Git GitLab Stata Google Analytics Salesforce Marketing Cloud.

PROFESSIONAL EXPERIENCE

EliteUS Software Solutions - Fedway, Edison, NJ

July 2024 – present

Client - Fedway Associates

Role - Artificial Intelligence Engineer

Developing an **AI chatbot** for Fedway system instructions using **NLP and AI**, integrating **text and image data processing**. Leveraged **Language models** from **AWS Bedrock**, **Azure** for **AI deployment** and **Streamlit** for dynamic interaction.

- Developed a Streamlit-based prototype chatbot integrated with Chroma vector databases, leveraging PyMuPDF for PDF parsing and Hugging Face Sentence Transformers for scalable NLP embeddings.
- Designed and deployed a Multi-Agent AI Bot trained on extensive help desk documentation using LLMs, Retrieval-Augmented Generation (RAG), and dynamic tool-calling for intelligent query resolution.
- Leading development of a Computer Vision product assistant using Azure Florence to automatically retrieve and display relevant product images, improving user engagement and task completion.
- Built an intelligent query-routing engine that classifies and maps user intents to the correct backend service using clustering, embeddings, and workflow policies.
- Designed and implemented a SQL routing engine for invoice-related queries, leveraging LLMs to translate natural language into optimized SQL, reducing manual support load.
- Designed an AutoGen-based Multi-Agent chatbot framework enabling autonomous task planning, workflow automation, query validation, and agent collaboration.
- Conducting research into agentic AI architectures to enhance autonomous decision-making, intent disambiguation, and multi-step task orchestration in enterprise chat systems.
- Managed OneDev Git repositories for CI/CD, version control, and cross-team collaboration, ensuring clean development workflows and rapid iteration cycles.
- Implemented HAProxy security configurations to enhance API reliability, enforce access control, and provide secure request routing for LLM-based services.
- Deployed LLaMA 3.1 and other models via AWS Bedrock for scalable inference; integrated Pillow for image processing and content formatting.
- Built a text-image similarity engine using the CLIP model to compare embeddings and auto-match product instructions with relevant images, improving multimedia search accuracy.
- Applied NumPy-based similarity matrices and cosine distance functions to rank and display best-matching image-text pairs in real time.

- Tech Stack: Python, AWS, Azure, LLaMA, OpenAI, Hugging Face, AutoGen, VectorDB (Chroma), PostgreSQL, CLIP, Florence, RAG, Multi-Agent Systems, CI/CD, HAProxy, Streamlit.

Data Science Intern, Harvest Software Solutions, LLC, Jacksonville, Florida

June 2023 – Sept 2023

Developed and implemented strategies using large-scale datasets for **optimizing real-time ad campaigns** which involved leveraging **H2O's AutoML, XGBoost, K-fold cross validation** to enhance performance by 70% and predictive accuracy by 40%. Utilized Aniview **APIs for real-time data extraction** and managed a dataset of over 5 million rows. Conducted **Exploratory Data Analysis (EDA) using python**, integrating CPM metrics for detailed performance evaluation and actionable insights.

- Employed **hypothesis testing, statistical analysis** to gain insights for data-driven decisions, optimize ad spend.
- Developed **predictive models** to minimize error, using cross-validation to prevent overfitting.
- **Deployed models into production** using **FastAPI** for predictions & used H2O's AutoML for model selection.
- Applied advanced natural language processing techniques using **Hugging Face Transformers** to analyze textual data from ad campaigns for **sentiment analysis** and **topic modeling** to understand customer engagement.
- **Tech Stack:** Python, Data Analysis, Hypothesis Testing, Machine Learning, Predictive Modeling, Model Deployment, NLP.

Database Analyst, Accenture Solutions Pvt. Ltd., Hyderabad, India

June 2021 – Aug 2022

Optimized **ETL processes** to cut support calls by 50%, and created **automated data pipelines with python**, boosting **efficiency by 40%**. Streamlined **data warehousing and DDL updates**, ensuring seamless data integration, performance with **SQL and Tableau**.

- Utilized **Amazon Athena and Redshift** for efficient querying and analysis of datasets over 50 million rows, leveraging serverless SQL capabilities and **scalable data warehousing**.
- Optimized **ETL workflows**, reducing support calls by 50%, enhancing data processing efficiency by 40% through **automated pipelines**. Optimized **DDL updates**, improving data management & system reliability by 15%.
- Conducted **data analysis** with **Tableau, PowerBI**, facilitating actionable insights and business intelligence.
- Designed & executed complex **SQL** queries using SQL Workbench for detailed **data manipulation** and analysis.
- Coordinated cross-functional teams for seamless **deployments on production servers**, ensuring data integration.
- Enhanced system performance through rigorous **testing, tuning, and monitoring** to ensure reliable operations in production.
- **Tech Stack:** Python, AWS(Athena, Redshift, S3, Sagemaker), ETL, SQL, Tableau, Data Warehousing, JIRA.

Research Assistant, Andhra University, Visakhapatnam, India

Jan 2021 – May 2021

Contributed to the creation and deployment of self-driving systems with autonomous navigation, lane tracking, and obstacle detection capabilities, emphasizing improvements in driving intelligence through **deep learning and computer vision** methods.

- Developed and optimized **Convolutional Neural Networks** for lane detection and object recognition. Leveraged **YOLO & Faster R-CNN** to enhance accuracy, speed of real-time object detection and classification.
- Applied image preprocessing techniques like **Gaussian blurring and color space conversion**, along with data augmentation, to enhance dataset quality and improve model generalization across various driving scenarios.
- Applied the **Hough Transform** for precise lane detection and vanishing point tracking to estimate road curvature, enhancing vehicle navigation. Integrated **multi-sensor data** from radar, lidar, and cameras to improve neural network accuracy.
- Applied **Neural Network Regression** for precise vehicle positioning and localization. Developed a real-time simulation environment using **Unity3D** with a **Flask backend** for dynamic interaction.
- **Tech Stack:** Python, Neural Networks, Computer Vision, Image Processing, Data Augmentation, Flask.

EDUCATION

Master of Science in Data Science (Statistics), Rutgers University, New Brunswick, NJ **Aug 2022 - May 2024**
Coursework: Statistical Modeling & Computing, Statistical Learning, Probability and Statistical Inference, Neural Networks, Machine Learning, Data Structures & Algorithms(Python), Data Wrangling (R), NLP, Data Mining.

School of Arts and Sciences | Research Assistant | Statistics Dept. **July 2024 - Present**

- Extracted & aggregated data from various lawyer websites to build a dataset, used for machine learning model development.
- Implemented machine learning models on the extracted data and developed efficient pipelines for automated data processing and integration. This approach optimized data flow and enhanced the overall performance of the machine learning systems.
- **Skills:** Python, BeautifulSoup, Machine Learning, Puppeteer Node.js, Data Wrangling, Pandas, RESTful APIs.

Bachelor of Technology in Electrical and Computer Engineering, Andhra University, India **June 2017 - June 2021**

PROJECTS

[Bayesian Networks, PyMC3, Informative Prior Distributions] Developed Bayesian logistic regression models using PyMC3 to predict diabetes status, incorporating informative priors. Evaluated performance with recall and f1-score metrics, comparing outcomes between uniform and normal distribution priors. [[Bayesian Classification on diabetes data](#)]

[Hugging Face transformers, Naive Bayes, Logistic Regression, BERT, N-Grams, GloVe embeddings] Developed using Naive Bayes, Logistic Regression, and DistilBERT, achieving 89% baseline accuracy globally. Enhanced precision to 94% with DistilBERT, incorporating n-grams analysis. [[Language Identification](#)]

[ML models - Random Forests, KNN, Logistic Regression, SVM, Decision Tree, and XGBoost] Executed diverse ML strategies and time series analysis for auto insurance fraud, prioritizing reduced false negatives. Chose K-Nearest Neighbors (KNN) with an impressive 97% recall. [[Auto Insurance Fraud Detection](#)]

[PostgreSQL, MongoDB, Python, Sorting techniques, Search Engine Optimization, Django] Created a user query-responsive tweet retrieval system by integrating dual databases: Postgres and MongoDB. Optimized data storage and retrieval efficiency by 35% while implementing advanced caching strategies for superior system performance. [[Twitter Search Application](#)]

LEADERSHIP

- Served as VP Membership at Toastmasters International (2018-2021). Analyzed trends to improve retention, led membership drives, and implemented programs to enhance participation and inclusivity.
- Mentored a candidate at Accenture, providing guidance, sharing expertise, and fostering their professional growth, resulting in enhanced technical skills and career advancement.
- Led cross-functional teams for seamless project delivery, managed end-to-end deployments to ensure timely and efficient project delivery, and resolved conflicts to maintain a productive environment.

